AP BIOLOGY TOPIC 8.4

Effect of Density of Populations & Carrying Capacity

UNIT 8: ECOLOGY (10–15% EXAM WEIGHT)

SYI-1.H | CARRYING CAPACITY (K) | DENSITY-DEPENDENT VS. INDEPENDENT

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 8.4 (EFFECT OF DENSITY OF POPULATIONS & CARRYING CAPACITY)

CED Core Vocabulary

Abiotic Limiting Factors — Sunlight, water availability, temperature, soil mineral nutrients (N, P), dissolved oxygen, and nesting/shelter space.

Biotic Limiting Factors — Food availability (prey/producers), interspecific competitors, predators, and pathogens.

Step 1 (Prey Boom) — When lynx numbers are low and winter food plants are abundant, the hare population grows rapidly.

Step 2 (Predator Lag Peak) — Abundant hares provide high food energy for lynx → lynx survival & litter sizes increase, causing the lynx peak to lag slightly behind the hare peak .

Step 3 (Prey Crash) — Heavy lynx predation + winter overbrowsing of willow shoots causes the hare population to crash.

Step 4 (Predator Crash) — Starvation causes the lynx population to crash shortly after, resetting the ~10-year cycle!

§2. 1. CARRYING CAPACITY (K) IS DYNAMIC, NOT FIXED

Resource Limits

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

1. Carrying Capacity (K) Is Dynamic, Not Fixed ➔ **Carrying Capacity (K)** ➔ **What Determines K?**

- **Carrying Capacity (K):** The maximum population size of a species that a specific environment can sustain indefinitely given available resources.
- **What Determines K ?**
  - **Abiotic Limiting Factors:** Sunlight, water availability, temperature, soil mineral nutrients (**N, P**), dissolved oxygen, and nesting/shelter space.
  - **Biotic Limiting Factors:** Food availability (prey/producers), interspecific competitors, predators, and pathogens.
- **Exceeding Carrying Capacity (Overshoot & Dieback):**
  - If a population rises above **K** (**N > K**), resources become severely depleted → **death rate (D) increases and birth rate (B) decreases** → negative feedback drives the population back down toward **K** .

[EXAMPLE & BIOLOGICAL APPLICATION] **Environmental Shifts Change K :** A multi-year drought reduces grass primary productivity (**NPP↓**), which directly lowers the carrying capacity (**K**) for herbivorous elk and bison in that ecosystem.

[COMMON AP EXAM FRQ TRAP] **Carrying capacity does NOT mean a population stays perfectly flat!** Real populations constantly **fluctuate / oscillate around** **K** due to seasonal resource changes and time lags in reproductive feedback.

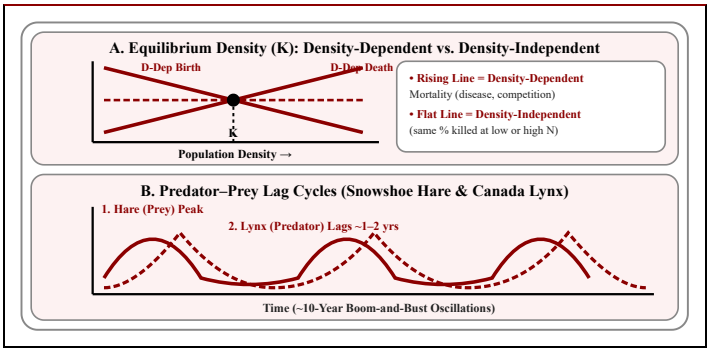
§3. 2. DENSITY-DEPENDENT VS. DENSITY-INDEPENDENT FACTORS

Population Regulation

FEATURE	DENSITY-DEPENDENT FACTORS	DENSITY-INDEPENDENT FACTORS
Definition	Limiting factors whose intensity/mortality rate **increases as population density increases** (negative feedback regulating **N** around **K**).	Limiting factors that reduce population size by the **same percentage regardless of population density** (cannot regulate around **K**).
Usual Nature	**Biotic interactions**	**Abiotic weather / catastrophic disturbances**
Key Mechanisms & Examples	• **Intraspecific Competition** for food, water, nutrients. • **Infectious Disease & Parasitism** (spreads faster in crowded hosts, e.g., flu, white-nose syndrome in bats). • **Predation** (predators switch to abundant prey). • **Territoriality & Nesting Space**. • **Metabolic Toxic Waste Buildup** (e.g., ethanol accumulation killing yeast at ~13–15%).	• **Severe Droughts & Floods** • **Wildfires & Volcanic Eruptions** • **Hurricanes, Tornadoes & Blizzards** • **Unseasonal Hard Freezes / Heatwaves** • **Human Habitat Destruction / Clear-Cutting / Pesticide Spraying**

§4. 3. VISUALIZING DENSITY REGULATION & PREDATOR–PREY CYCLES

Snowshoe Hare & Lynx



- **Predator–Prey Boom-and-Bust Cycles (Snowshoe Hare & Canada Lynx):**
  - **Step 1 (Prey Boom):** When lynx numbers are low and winter food plants are abundant, the hare population grows rapidly.
  - **Step 2 (Predator Lag Peak):** Abundant hares provide high food energy for lynx → lynx survival & litter sizes increase, causing the **lynx peak to lag slightly behind the hare peak**.
  - **Step 3 (Prey Crash):** Heavy lynx predation + winter overbrowsing of willow shoots causes the hare population to crash.
  - **Step 4 (Predator Crash):** Starvation causes the lynx population to crash shortly after, resetting the ~10-year cycle!

[AP EXAM FRQ SCORING STRATEGY] **Graph Reading Tip for Predator–Prey FRQs:** The **prey population almost always has a higher population size** (due to the 10% energy rule) and **peaks FIRST**; the predator curve has a lower peak and **shifts to the right (time lag)**!

§5. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 8.4 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 8.4: Effect of Density of Populations & Carrying Capacity ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 10, 11 & 12 (Energy Dynamics, Whole-Plant Transpiration & Pillbug Animal Behavior Choice Chambers): Choice chambers testing pillbug (Armadillidium vulgare) chemotaxis/hydrotaxis/kinesis (count pillbugs on wet vs. dry filter paper every 30 s for 10 min; evaluate using χ^2 against 50:50 null distribution). Potometer/gravimetric transpiration across light, wind, and humidity treatments normalized per cm² leaf surface area. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Ecological Equations: Population Growth Rate: $dN/dt = B - D$. Exponential Growth: $dN/dt = r_{max} \cdot N$ (J-curve). Logistic Growth: $dN/dt = r_{max} \cdot N \cdot [(K - N)/K]$ (S-curve; maximum instantaneous growth rate dN/dt occurs at $N = K/2$). Simpson's Diversity Index: $D_I = 1 - \Sigma(n / N)^2$ (ranges from 0 to 1; higher value = higher diversity). Net Primary Productivity: $NPP = GPP - R$. Trophic 10% Rule: ~10% of energy transfers to the next trophic level (~90% lost as metabolic heat/respiration). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Ecological Disruption & Trophic Cascade Case Study: Removal of a keystone predator (e.g., Pisaster ochraceus sea star in rocky intertidal zones, or sea otters in kelp forests) triggers a top-down trophic cascade: prey populations (mussels or sea urchins) explode unchecked, competitively excluding other invertebrates or overgrazing kelp producers, collapsing Simpson's Diversity Index. Similarly, agricultural N/P fertilizer runoff causes Cultural Eutrophication: algal blooms block light, aerobic bacteria decompose dead algae and deplete dissolved O_2, creating hypoxic dead zones.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 8.4 to prove why the claim holds.

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